Faculty of Engineering — Shoubra
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FACULTY OF ENGINEERING AT SHOUBRA

Database Feature Tables

SW Phase (1)

EdgeAI Location Intelligence

FPGA-Based Cloud-Integrated AI System for Intelligent Location Analytics

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Database Schema Documentation

Overview

The **EdgeAI Location Intelligence** database is designed to integrate data collected from FPGA-based edge devices, business analytics layers, and cloud-hosted AI models. It supports structured data ingestion, aggregation, and predictive insights generation for intelligent location-based decision systems.

Schema Overview

The database schema consists of multiple interlinked tables organized under the `public` schema. Each table focuses on a specific part of the system: locations, business metadata, real-time edge data, financial data, or AI model predictions.

- **Location Metadata:** Defines each monitored geographic or business area.
- **Business Profile:** Contains business-level operational and financial descriptors.
- **Real-Time Metrics:** Holds sensor-level data from FPGA/Edge AI units.
- **Daily FPGA Metrics:** Aggregated daily summaries derived from real-time data.
- **Daily Revenue:** Financial data used for AI training and evaluation.
- **Prediction Output:** Stores ML predictions and confidence metrics.
- **Weather & Events:** External contextual factors influencing performance.
- **Promotions:** Marketing campaigns and their expected impacts.

1 Location_Metadata Table

Purpose

Stores static metadata about monitored locations — including geographic, demographic, and competitive features — which serve as base context for analytics and business association.

Structure: Location_Metadata

Column	Type	Description	Key
location_id	SERIAL	Unique location identifier	PK
latitude	DOUBLE PRECISION	Geographic coordinate (latitude)	
longitude	DOUBLE PRECISION	Geographic coordinate (longitude)	
location_name	TEXT	Optional readable name	
location_type	TEXT	Type of monitored area (mall, market, etc.)	
population	INTEGER	Estimated population density around the area	
avg_income	NUMERIC(12,2)	Average local income level	
competitor_count	INTEGER	Number of nearby competitors	
catchment_radius_m	INTEGER	Analysis radius around location in meters	
created_at	TIMESTAMP	Record creation timestamp	Default now()

Feature Importance Notes

- **Population:** Often correlates strongly with daily footfall and total revenue.
- **Avg_Income:** Useful for predicting purchasing power and spending habits.
- **Competitor_Count:** Negatively impacts profitability potential in dense zones.
- **Catchment_Radius_m:** Adjusts for different business attraction areas.

Relationships

- One-to-Many with `Business_Profile`.
- One-to-Many with `RealTime_Metrics`, `Daily_FPGA_Metrics`, and `Weather`.
- Connected to `Prediction_Output` via foreign key.

2 Business_Profile Table

Purpose

Captures detailed business attributes linked to each monitored location, including operational capacity, resource costs, and owner experience, forming a key input for profit/revenue models.

Structure: Business_Profile

Column	Type	Description	Key
business_id	SERIAL	Unique business identifier	PK
location_id	INTEGER	Linked to `Location_Metadata`	FK
name	TEXT	Registered business name	
business_type	TEXT	Type of business (restaurant, retail, etc.)	
product_type	TEXT	Primary product or service type	
owner_experience_years	INTEGER	Experience level of owner/manager	
work_hours_per_day	INTEGER	Average operating hours per day	
work_days_per_week	INTEGER	Working days per week	
store_area_m2	NUMERIC	Physical area of the business	
rent_monthly	NUMERIC(12,2)	Monthly rent cost	
staff_cost_monthly	NUMERIC(12,2)	Monthly staff expense	
utilities_cost_monthly	NUMERIC(12,2)	Electricity, water, internet, etc.	
opening_date	DATE	Date when the business started	
created_at	TIMESTAMP	Record creation timestamp	Default now()

Feature Importance Notes

- **Rent_Monthly, Staff_Cost, Utilities_Cost:** Strong indicators of operational load and net profitability.
- **Owner Experience:** Often impacts performance variance in early-stage businesses.
- **Business Type & Product Type:** Key categorical features for model segmentation.

Relationships

- One-to-One with `Location_Metadata`.
- One-to-Many with `Daily_Revenue`, `Prediction_Output`, and `Promotions`.

3 RealTime_Metrics Table

Purpose

Stores high-frequency sensor data collected from FPGA or edge AI systems deployed at various locations. These metrics are crucial for real-time monitoring, crowd analytics, and short-term forecasting.

Structure: RealTime_Metrics

Column	Type	Description	Key
metric_id	BIGSERIAL	Unique metric record ID	PK
location_id	INTEGER	Linked to `Location_Metadata`	FK
timestamp	TIMESTAMP	Time of data capture from FPGA node	
people_count	INTEGER	Number of detected individuals in frame/zone	
dwell_time_avg	NUMERIC	Average time spent per person (seconds)	
vehicle_count	INTEGER	Number of vehicles detected (if applicable)	
data_quality_flag	SMALLINT	Indicates missing/corrupted data quality	Default 0
created_at	TIMESTAMP	Record creation timestamp	Default now()

Feature Importance Notes

- **People_Count:** Directly correlates with potential footfall and sales activity.
- **Dwell_Time_Avg:** Indicates engagement or customer interest in location.
- **Vehicle_Count:** Proxy for accessibility and commercial exposure.
- **Data_Quality_Flag:** Ensures AI models ignore unreliable readings.

Relationships

- Many-to-One with `Location_Metadata`.
- Serves as source data for `Daily_FPGA_Metrics`.

4 Daily_FPGA_Metrics Table

Purpose

Contains aggregated daily summaries generated from `RealTime_Metrics`. These metrics provide daily-level insights that can be used for trend analysis, ML training, and anomaly detection.

Structure: Daily_FPGA_Metrics

Column	Type	Description	Key
location_id	INTEGER	Linked to `Location_Metadata`	PK, FK
date	DATE	Date of aggregation	PK
daily_footfall	INTEGER	Total daily visitor count	
daily_avg_dwell	NUMERIC	Average dwell time across all visitors	
daily_vehicle_count	INTEGER	Total number of detected vehicles per day	
hours_of_operation	INTEGER	Total hours monitored per day	
missing_intervals	INTEGER	Count of missing or faulty data segments	
created_at	TIMESTAMP	Record creation timestamp	Default now()

Feature Importance Notes

- **Daily_Footfall:** Key behavioral metric used for sales and resource forecasting.
- **Missing_Intervals:** Affects accuracy of subsequent revenue predictions.
- **Daily_Avg_Dwell:** Helps quantify engagement trends over time.

Relationships

- Many-to-One with `Location_Metadata`.
- Serves as historical data input for prediction models.

5 Daily_Revenue Table

Purpose

Stores financial data representing total daily revenue and transactions per business. This table serves as the ground truth (label data) for supervised machine learning models.

Structure: Daily_Revenue

Column	Type	Description	Key
business_id	INTEGER	Linked to `Business_Profile`	PK, FK
date	DATE	Date of revenue record	PK
total_revenue	NUMERIC(14,2)	Total daily sales in local currency	
total_transactions	INTEGER	Total transactions made	
avg_transaction_value	NUMERIC(12,2)	Mean spending per transaction	

Feature Importance Notes

- **Total_Revenue:** Target variable for prediction models.
- **Total_Transactions:** Reflects customer activity density.
- **Avg_Transaction_Value:** Indicates customer purchasing power.

Relationships

- Many-to-One with `Business_Profile`.
- Supports time-series and regression analyses.

6 Prediction_Output Table

Purpose

Holds results of AI/ML prediction models, including predicted revenue, profit, and confidence scores for given business and location combinations.

Structure: Prediction_Output

Column	Type	Description	Key
prediction_id	BIGSERIAL	Unique prediction record	PK
business_id	INTEGER	Linked to `Business_Profile`	FK
location_id	INTEGER	Linked to `Location_Metadata`	FK
prediction_date	DATE	Date of prediction output	
predicted_revenue	NUMERIC(14,2)	Forecasted revenue	
predicted_profit	NUMERIC(14,2)	Forecasted net profit	
confidence_score	REAL	Model prediction confidence	
model_version	TEXT	Model version identifier	
created_at	TIMESTAMP	Record creation timestamp	Default now()

Feature Importance Notes

- **Predicted_Revenue & Profit:** Outputs from the regression model.
- **Confidence_Score:** Used for decision reliability visualization.
- **Model_Version:** Ensures traceability across ML experiments.

Relationships

- Many-to-One with `Business_Profile` and `Location_Metadata`.
- Serves analytics dashboard visualization layer.

7 Weather Table

Purpose

Integrates external weather data to provide environmental context for model predictions and footfall variations.

Structure: Weather

Column	Type	Description	Key
weather_id	SERIAL	Unique weather record	PK
location_id	INTEGER	Linked to `Location_Metadata`	FK
timestamp	TIMESTAMP	Observation time	
temp_c	REAL	Temperature in Celsius	
precip_mm	REAL	Rainfall amount in millimeters	
is_rain	BOOLEAN	True if raining	
created_at	TIMESTAMP	Record creation timestamp	Default now()

8 Promotions Table

Purpose

Stores promotional campaigns for businesses to evaluate their effects on sales and footfall performance.

Structure: Promotions

Column	Type	Description	Key
promo_id	SERIAL	Unique promotion identifier	PK
business_id	INTEGER	Linked to `Business_Profile`	FK
start_date	DATE	Promotion start date	
end_date	DATE	Promotion end date	
promo_type	TEXT	Type of promotion (discount, event, etc.)	
expected_lift_pct	REAL	Estimated increase in revenue (%)	
created_at	TIMESTAMP	Record creation timestamp	Default now()

9 Events Table

Purpose

Records information about external events such as festivals or exhibitions that could impact traffic, footfall, or revenue near monitored areas.

Structure: Events

Column	Type	Description	Key
event_id	SERIAL	Unique event identifier	PK
name	TEXT	Event name or title	
location_lat	DOUBLE PRECISION	Latitude of event location	
location_lon	DOUBLE PRECISION	Longitude of event location	
start_date	DATE	Start date of event	
end_date	DATE	End date of event	
expected_attendance	INTEGER	Expected number of attendees	
created_at	TIMESTAMP	Record creation timestamp	Default now()

Feature Importance Notes

- **Expected_Attendance:** Key variable to estimate external impact on nearby businesses.
- **Location_Lat/Lon:** Used for geospatial joins with monitored sites.

Relationships

- Independent contextual table, indirectly linked by location proximity.